Lost&Found Telegram Bot — System Requirements Document

1. Project Overview

Project name: Lost&Found

System: Telegram chatbot and announcement channel

Target users: University students and staff

Purpose

The purpose of the Lost&Found bot is to provide a central and trusted place for university students and staff to report lost and found items. The system helps users find existing reports, communicate about item returns, and coordinate with Student Affairs when necessary.

Currently, lost-and-found announcements are scattered across Telegram groups, making items difficult to locate.

2. Functional Requirements

Functional requirements describe what the system must do.

FR-01: User access

- Students and staff must be able to use the bot through a Telegram account.
- Users must be able to start the bot and navigate its main functions.

FR-02: Found-item reporting

- Users must be able to submit a report when they find an item.
- The report must include a title, description, location, and date.
- Users must be able to attach an optional photo.
- The report must be published to the Telegram announcement channel with the `#FOUND` hashtag.
- Users must be able to indicate that the item has been handed to Student Affairs or that direct return is preferred.

FR-03: Lost-item reporting

- Users must be able to report items they have lost.
- The report must include a title, description, location, and date.
- Users must be able to attach an optional photo.
- The report must be published with the `#LOST` hashtag.

FR-04: Search and navigation

- Users must be able to search or browse `#FOUND` announcements to look for their missing items.
- The bot must provide inline buttons for easier navigation.
- The bot must provide a list of campus locations for tagging reports.

FR-05: Private contact requests

- Users must be able to request contact with the person who posted a found-item report.
- Contact requests must be handled through the bot rather than publishing the person's contact details in the public channel.
- The system must follow the project's contact-request procedure.

FR-06: Administration and moderation

- Administrators must be able to remove inappropriate or false posts using an administrative command.
- Ordinary users must not be allowed to perform administrator-only actions.
- Administrators must be able to review claim history and resolve disputes.

FR-07: Item return and support

- The system must support the process of returning found items to their owners.
- Items may be kept in the Student Affairs room or returned directly to the owner when appropriate.
- The responsible support staff must help coordinate returns and resolve cases requiring assistance.

3. Non-Functional Requirements

Non-functional requirements describe how the system must operate.

NFR-01: Timeliness

- Eligible reports submitted between **09:00 and 22:00** must appear in the Telegram channel within two hours.
- Reports submitted before or after this time window must appear the next day.

NFR-02: Privacy

- Personal contact details must not be exposed in public channel announcements.
- Contact requests must be handled privately through the bot according to the defined procedure.

NFR-03: Security and access control

- Administrative commands must be restricted to authorized administrators.
- The system must prevent ordinary users from removing reports or accessing administrative functions.

NFR-04: Usability

- The bot must provide simple navigation using Telegram commands, inline buttons, and campus-location options.
- Reporting an item should follow a clear, step-by-step process.

NFR-05: Reliability

- Valid reports must be processed and published according to the required schedule.
- Errors during submission or publication should be handled without unintentionally creating duplicate reports.

NFR-06: Maintainability

- The system should be organized so that developers can update reporting flows, location options, and administrative functions without rebuilding the entire bot.

NFR-07: Traceability

- Claim history must be available to administrators for reviewing cases and resolving disputes.

4. Acceptance Criteria

The system will be considered ready for acceptance when:

1. A student or staff member can submit a valid FOUND report.
2. A user can submit a valid LOST report.

3. Reports contain the required title, description, location, date, and optional photo.
4. Users can search or browse found-item announcements and use the navigation buttons.
5. Publication follows the specified time window and deadline.
6. Contact requests are handled privately.
7. Administrators can remove inappropriate posts and review claim history.
8. Administrators can mark disputes as resolved.
9. Users can follow the documented process for returning items through Student Affairs or directly to the owner.

5. Project Roles

- **Project Manager:** Coordinates planning, requirements, schedule, and final documentation.
- **Tester & User Researcher:** Collects user feedback and verifies acceptance criteria.
- **Developer:** Implements the Telegram bot and its database and workflows.
- **Support & Student Affairs Mediator:** Defines and coordinates the item-return and support procedures.
- **Administrator & Moderator:** Handles inappropriate posts, reviews claims, and supports dispute resolution.

6. Scope Notes

This document summarizes the requirements in the project specification. Detailed technical choices, such as database structure, hosting, backup strategy, and exact search implementation, should be documented separately if they are required for the final system.